Russell Dudek

deephow INDEPENDENT CANDIDATE

PHYSICAL AI PLATFORM STRATEGY | INDUSTRIAL OPERATIONS | AI PRODUCT JUDGMENT

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ROLE-ALIGNED PROFILE

Operator-technologist who makes complex work visible and turns emerging technology into adopted operating mechanisms. Brings direct experience across software-enabled robotics, manufacturing, quality, telemetry, enterprise workflows, autonomous systems, AI readiness, and cross-functional technical execution. Particularly effective at connecting factory-floor realities, AI reliability, platform decisions, customer outcomes, and organizational adoption.

EXPERIENCE

Vape-Jet · Director of Operations — 2025–Present

- Leads AI-first operating transformation across production, purchasing, quality, customer success, support, engineering issue flow, ERP/Odoo, Jira, HubSpot, Slack, telemetry, and AI-assisted workflows.
- Connects factory, field, customer, and machine-fleet signals into prioritized work, earlier risk detection, cleaner handoffs, dashboards, standard work, training, and accountability.
- Works directly in a software-enabled robotics and machine-vision environment with approximately 1,000 fielded robots informing support and operating feedback loops.

DudeWorth · Founder / AI and Automation Advisor — 2017–Present

- Develops AI-readiness and opportunity frameworks assessing value, feasibility, data readiness, governance, adoption burden, reuse, and time-to-value.
- Designs agentic workflow patterns using structured context, retrieval, AI agents, human judgment, escalation, decision records, evaluation, and governance rails.
- Advises on AI-enabled operations, logistics technology, warehouse automation, AMRs, ASRS, cobots, transportation systems, network analysis, and workflow design.

Amazon · Operations Management / Site Leader — 2014–2018

- Led launch and execution for Amazon Prime Pittsburgh in a 24/7 customer-backward environment supporting approximately five million second-day orders annually.
- Built leadership routines, labor planning, throughput visibility, escalation paths, standard work, and operating mechanisms across safety, quality, flow, customer experience, and engagement.
- Served on the Amazon Prime Now launch team with Whole Foods and applied Lean and Gemba practices across operating priorities.
- Contributed to logistics innovation through Prime Air research and Amazon Flex geolocation concepts.

IMMEDIATE CONTRIBUTION

PLATFORM JUDGMENT

Work-object definition, standardize/federate decisions, roadmap tradeoffs, technical debt, reuse, and first-value compression.

PHYSICAL AI CONTEXT

Robotics, machine vision, autonomous systems, factory workflows, quality, telemetry, standard work, and human actuation.

AI OPERATING CONTRACTS

Data readiness, confidence, evaluation, human authority, escalation, governance, monitoring, and learning loops.

CROSS-FUNCTIONAL EXECUTION

Engineering, operations, quality, support, customers, business systems, GTM translation, and operating cadence.

SELECTED EVIDENCE AGAINST THE MANDATE

PHYSICAL SYSTEMS

Robotics, machine vision, autonomous systems, factory workflows, field operations, and machine-fleet telemetry.

SCALE AND ACTIVATION

Prime Pittsburgh launch and operating mechanisms supporting approximately five million second-day orders annually.

TRUST AND RELIABILITY

Quality systems, root cause, process control, high-consequence technical delivery, and regulated operations.

AI PRODUCT MECHANISMS

Readiness, RAG and agentic workflows, human judgment, evaluation, governance, escalation, and learning loops.

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EXPERIENCE CONTINUED

Compunetics · Director of Operations and Engineering Management 2000–2013

- Progressed through R&D program management, production management, strategic acquisition, operations management, and director-level leadership.
- Led engineering, manufacturing, quality, customer delivery, and complex technical programs in advanced electronics and high-reliability environments.
- Built quality systems, standard work, root-cause practices, performance measures, and process controls; collaborated with DoD, CERN, Intel, and AMD.
- Published industry articles, delivered technical talks, and participated in the HyperTransport Consortium.

ZeusVu · Chief Pilot, Autonomous Systems and Aerial Data Operations 2018–2022

- Led regulated drone operations across medical, energy, logistics, real-estate, and restricted-airspace contexts with a 100% safety record and no FAA incidents.
- Developed TensorFlow-based concepts for automated aerial inventory and computer-vision-enabled field intelligence.
- Designed medical drone delivery and chain-of-custody concepts connecting technology, risk, stakeholders, and governed execution.

Cardinal Building Products · Regional Director 2021–2023

- Built the company's first remote regional operation and modernized distributed customer, CRM, commercial, fulfillment, logistics, digital marketing, SEO, and online-buying workflows.
- Advanced AI-enhanced transportation-management and route-planning concepts.

ADDITIONAL LEADERSHIP

Enviro Social Capital · Chairman of the Board, 2014–2017: Co-founded and led a public-private impact initiative focused on sustainable infrastructure, green finance, and civic transformation; helped shape and evaluate a $200 million Social Impact Bond feasibility concept for Pittsburgh green infrastructure.

IEEE · CMPT and EDS Pittsburgh Chapter Chair, 2012–2016: Led regional programming connecting industry, academia, STEM, electronics, manufacturing, and emerging technology; organized conferences, seminars, business tours, and technical programming.

TECHNICAL TOOLKIT

Agentic AI · LLM workflows · RAG patterns · Human-in-the-loop design · Python · SQL · JavaScript · TensorFlow · Tableau · ERP/Odoo workflows · Jira · Confluence · HubSpot · Slack · GIS and mapping workflows

DEEPHOW PLATFORM FIT

KNOWLEDGE → EXECUTION

Experience translating expert practice into standard work, training, visible controls, and operating routines.

EXECUTION → EVIDENCE

Quality systems, root cause, process control, telemetry, issue flow, and high-consequence technical delivery.

EVIDENCE → IMPROVEMENT

Lean/Gemba mechanisms, performance visibility, escalation, prioritization, and repeatable learning.

PLATFORM → ADOPTION

Workflow design, training, operating cadence, customer translation, and mechanisms that make new technology usable in daily work.

EDUCATION

University of Pittsburgh · Bachelor of Science
STEM grounding includes materials science, physical chemistry, and biology-related studies. Academic and professional publications are indexed on Google Scholar.

CREDENTIALS

Google AI Essentials
Food Safety Management Certification
IBM Enterprise Design Thinking Practitioner
Six Sigma Certification
OSHA 10

OPERATING PRINCIPLES

Start with the customer outcome. Make decision rights visible. Pilot before institutionalizing. Set AI thresholds by consequence. Preserve human authority. Turn deployments into reusable platform learning.